

Supplementary Information

CrossPiezo: Elite-Tail Instability in Cross-Database Piezoelectric Screening

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1 Supplementary notes

1.1 Portfolio estimand definition (Note 1)

The portfolio benchmark reports two related quantities. *Worst-source recall* is a material-level (micro) metric: it counts how many of the top- q^*n materials selected by a strategy are also present in the JARVIS and MP elite sets, divided by q^*n , with no group weighting. *Paired difference in recall* is the difference between the material-level worst-source recall of the strategy and that of the better single-source baseline, computed on the full observed panel.

The confidence interval reported in Table S3 uses a *full-procedure grouped bootstrap*. Each replicate: (i) resamples reduced-formula groups with replacement and assigns distinct bootstrap identities to duplicated occurrences; (ii) recomputes source-specific rankings and top- q^* elite sets on the bootstrap sample; (iii) re-runs the portfolio strategy and both single-source baselines; (iv) computes the improvement over the better single source within that replicate. The reported 95% interval is the percentile bootstrap interval of these replicate differences. The point estimate is fixed on the full-panel selections and is exactly the arithmetic difference of the Recall column; the bootstrap captures uncertainty from group resampling and re-selection.

The “better single source” is determined per replicate for the interval; the main-text point estimate uses the full-panel better source (here JARVIS-only and MP-only are tied on P0 F1 at $q^* = 10\%$, $b = 1.0$). The portfolio analysis is presented as a risk-management illustration, not as a validated method comparison or as evidence of physical truth.

1.2 Structure matching and manual audit (Note 2)

The CrossPiezo-Invariant-v1 panel was constructed from relaxed CIFs shared by the Materials Project (MP) and JARVIS. The matching pipeline is: (i) load all processed MP and JARVIS piezoelectric records; (ii) compute structure-level overlaps by compositional reduced formula and relaxed-lattice similarity; (iii) apply a frozen similarity threshold to obtain candidate pairs; (iv) perform a response-blind manual audit of the top-matched candidates using a frozen rule set. The audit was response-blind: the auditor did not see piezoelectric response values when deciding whether a candidate pair represented the same or a distinct structure. All audit decisions, exclusion reasons and ambiguous cases are recorded in the frozen panel-membership file. P0 ($n = 573$) is the full set of accepted structure matches; P2 ($n = 207$) is the subset passing a tighter similarity threshold.

1.3 Tensor invariants and conventions (Note 3)

All piezoelectric response values are piezoelectric stress tensors $\mathbf{e}$ in C/m^2 with full Cartesian $3 \times 3 \times 3$ components. The three coordinate-invariant scalars are:

- F1, Cartesian Frobenius norm $\|\mathbf{e}\|_F$;
- F3, true collinear longitudinal maximum $\max_{\|\mathbf{n}\|=1} |\mathbf{n}_i e_{ijk} \mathbf{n}_j \mathbf{n}_k|$ computed by numerical optimisation over the unit sphere;
- F4, Kelvin/Mandel operator norm on symmetric strain, computed from the Mandel representation of the 6×3 piezoelectric tensor.

Voigt-to-Cartesian conversion follows the standard convention $11 \rightarrow xx$, $22 \rightarrow yy$, $33 \rightarrow zz$, $23 \rightarrow yz$, $13 \rightarrow xz$, $12 \rightarrow xy$ with factor-of-two handling for shear components as appropriate for the stress tensor. No sign conventions, unit conversions or coordinate rotations were applied silently; all transformations are recorded in the repository transformation history.

1.4 Bootstrap, quantile and tie handling (Note 4)

Screening-resolution curves are evaluated at quantiles $q = 1\%, 2\%, \dots, 50\%$. The chance-adjusted Jaccard overlap is $\tilde{J}_q = (J_q - \mathbb{E}[J_q]) / (1 - \mathbb{E}[J_q])$, where $\mathbb{E}[J_q]$ is the exact hypergeometric expectation under independent rankings. Simultaneous 95% confidence bands for the screening-resolution curve use a paired bootstrap with a studentized sup-norm critical value; this bootstrap resamples matched material pairs (rows) because the curve itself is a cross-sectional function of the ranked universe and the band is constructed for fixed quantiles.

For property-control comparisons and the portfolio benchmark, bootstrap resampling is by reduced-formula group to respect the dependence among entries sharing a composition. In the portfolio full-procedure bootstrap, duplicated groups create distinct bootstrap identities so that re-selection and elite-set sizes are computed on the expanded bootstrap sample. The random seed, number of replicates, studentization details, tie handling and k rounding are recorded in the Phase 7C configuration and result files. Ties in source rankings are broken by stable sorting; duplicate reduced formulae are retained in the ranking but noted in the panel metadata. The persistent-onset criterion requires the lower confidence bound to exceed a threshold δ for at least five consecutive quantiles.

2 Supplementary figures

3 Supplementary tables

Table 1: Corrected high-response sensitivity for P0 F1 at $f = 0.10$ and $f = 0.25$.

Method & f & N & au & CI & nAUCC & CI						
Dual-high	10%	58	0.035	[-0.193, 0.239]	-0.046	[-0.118, 0.044]
JARVIS anchor	10%	57	0.209	[-0.062, 0.374]	0.053	[-0.061, 0.191]
MP anchor	10%	57	-0.053	[-0.235, 0.137]	-0.018	[-0.104, 0.090]
Pooled (exploratory)	10%	58	—	[-, -]	-0.155	[-, -]
Dual-high	25%	143	0.070	[-0.046, 0.181]	0.031	[-0.034, 0.102]
JARVIS anchor	25%	143	-0.019	[-0.141, 0.105]	-0.006	[-0.070, 0.070]
MP anchor	25%	143	0.039	[-0.083, 0.154]	0.007	[-0.052, 0.078]
Pooled (exploratory)	25%	144	—	[-, -]	-0.100	[-, -]

Table 2: Property-control comparison on P0. The same-field copy flag is raised when more than 5% of finite matched pairs have identical JARVIS/MP values.

Attribute & N & au & 95% CI & nAUCC & 95% CI & $\Delta\tau$ & Δ nAUCC & Copy flag									
Volume	& 573	& 0.967	& [0.963, 0.971]	& 0.918	& [0.901, 0.936]	& 0.722	& 0.813	&	
Band gap	& 573	& 0.908	& [0.890, 0.923]	& 0.897	& [0.864, 0.908]	& 0.663	& 0.792	&	
Energy above hull	& 573	& 0.611	& [0.547, 0.670]	& 0.533	& [0.455, 0.592]	& 0.366	& 0.428	&	✓
Dielectric trace	& 573	& 0.482	& [0.418, 0.547]	& 0.286	& [0.228, 0.344]	& 0.236	& 0.181	&	

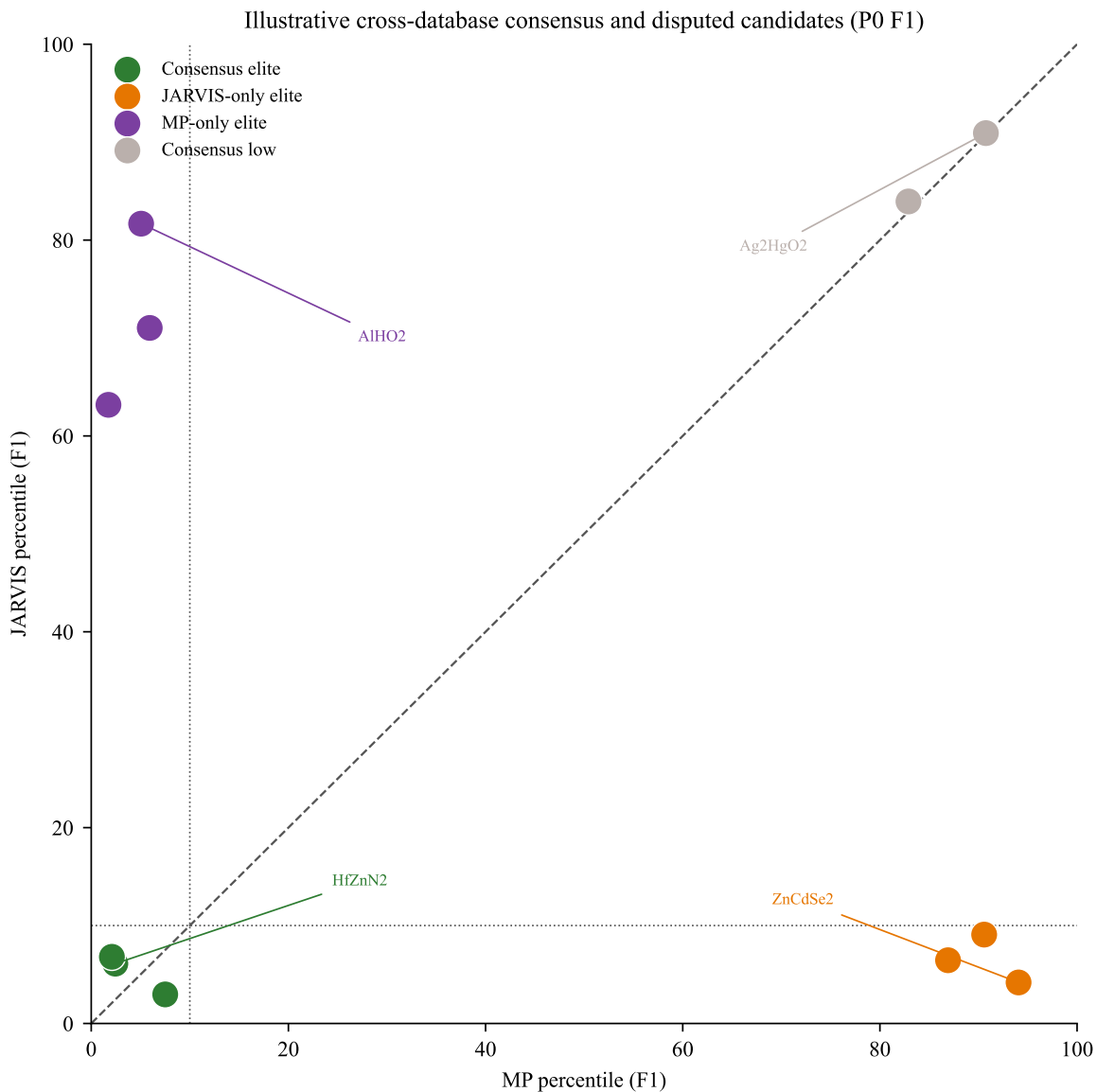


Figure 1: Illustrative cross-database consensus and disputed candidates on P0 F1. Points show JARVIS versus MP percentiles for consensus elite (green), JARVIS-only elite (orange), MP-only elite (purple) and consensus low (grey) materials. The dashed diagonal marks equal ranking; dotted lines mark the 10% elite threshold. These examples are illustrative and are not adjudicated as physically correct or incorrect.

Portfolio cost-coverage frontier on P0 F1 ($q^* = 10\%$)

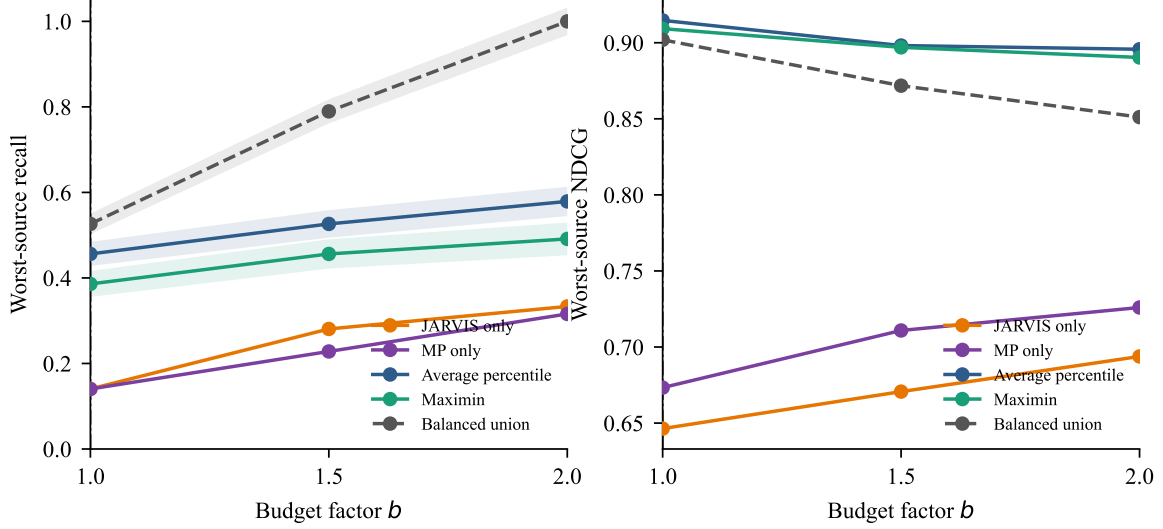


Figure 2: Portfolio cost-coverage frontier on P0 F1 ($q^* = 10\%$). Worst-source recall (left) and NDCG (right) are plotted against budget factor b for the best single-source baselines and three source-robust aggregation strategies. The vertical dashed line marks the equal-budget analysis at $b = 1.0$. Aggregation changes the coverage-quality trade-off but does not validate physical truth. See Note 1 for the distinction between material-level recall and group-mean paired differences.

Table 3: Equal-budget ($b = 1.0$) portfolio results on P0 F1 ($q^* = 10\%$). Paired differences are versus the better single-source baseline and are computed as material-level worst-source recall differences using a full-procedure grouped bootstrap: each replicate resamples reduced-formula groups with replacement, assigns distinct identities to duplicated occurrences, re-runs the portfolio strategy and both single-source baselines, and recomputes the improvement (see Note 1).

Strategy	& Recall	& NDCG	& Coverage	& Δ Recall	& 95% CI
JARVIS only	& 0.140	& 0.646	& 0.099	& 0.000	& [0.000, 0.000]
MP only	& 0.140	& 0.673	& 0.099	& 0.000	& [0.000, 0.000]
Average percentile	& 0.456	& 0.915	& 0.099	& 0.316	& [0.172, 0.362]
Borda	& 0.456	& 0.915	& 0.099	& 0.316	& [0.172, 0.362]
Maximin	& 0.386	& 0.909	& 0.099	& 0.246	& [0.107, 0.286]
Intersection first	& 0.456	& 0.915	& 0.099	& 0.316	& [0.172, 0.362]
Balanced union	& 0.526	& 0.902	& 0.099	& 0.386	& [0.273, 0.441]
Disagreement abstention	& 0.456	& 0.915	& 0.099	& 0.316	& [0.172, 0.362]
Minimax oracle	& 0.474	& 0.915	& 0.099	& 0.333	& [0.250, 0.414]

Table 4: Portfolio worst-source recall on P0 F1 across budget factors $b = 1.0, 1.5, 2.0$.

Strategy & $b = 1.0$ & $b = 1.5$ & $b = 2.0$
JARVIS only & 0.140 & 0.281 & 0.333
MP only & 0.140 & 0.228 & 0.316
Average percentile & 0.456 & 0.526 & 0.579
Maximin & 0.386 & 0.456 & 0.491
Balanced union & 0.526 & 0.789 & 1.000